EXPERIMENT-5b

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Date:

Simulate the Life Cycle Stages for UI Design Using the RAD Model and Develop a Small Interactive Interface Using OpenProj

AIM

The aim of this experiment is to simulate the lifecycle stages of UI design using the Rapid Application Development (RAD) model and to design a small interactive Student Dashboard interface. The project schedule is tracked and managed using OpenProj, an open-source project management tool.

Tool Used

- OpenProj – for project scheduling and Gantt chart creation
- Axure RP – for wireframing and interactive prototyping
- Figma – for high-fidelity visual design
- Tool Link: https://sourceforge.net/projects/openproj/

Procedure

Step 1: Requirements Planning

The first step involves identifying the core requirements for the Student Dashboard UI. The goal is to determine what information the dashboard must display and what actions a student user needs to perform.

1. Gather Requirements:

- Identify key features: Course overview, Attendance tracker, Assignment list, Notifications panel, and Profile section.
- Define the target user: A student who needs quick access to academic information and updates.
- Specify interface elements: Navigation sidebar, summary cards, progress bars, data tables, and notification badges.

2. Define Use Cases:

- Use Case 1: Student views today's class schedule and attendance percentage on the home dashboard.
- Use Case 2: Student checks pending assignments with due dates and marks them as completed.
- Use Case 3: Student views notifications for new announcements, grade updates, and upcoming deadlines.
- Use Case 4: Student navigates to the Profile section to update personal details.

Output in OpenProj:

- Created a new project titled "Student Dashboard UI – RAD Model".
- Added tasks: "Gather Requirements" (Duration: 2 days) and "Define Use Cases" (Duration: 2 days).

- Set dependencies and assigned resources to each task.

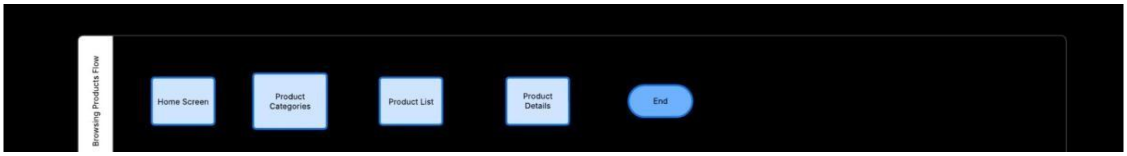

Fig 1: OpenProj – Requirements Planning Tasks

Step 2: User Design

In this step, initial sketches and digital wireframes are created for the Student Dashboard screens. The wireframes define the layout, navigation structure, and key UI components.

1. Sketch Initial Designs:

- Drew rough paper sketches showing the sidebar navigation, main content area, and widget placement.
- Planned a responsive layout with a collapsible sidebar for desktop and mobile views.

2. Create Digital Wireframes:

- Home Dashboard: Summary cards for Attendance %, Pending Assignments, Upcoming Classes, and GPA.
- Courses Page: List of enrolled courses with progress bars showing completion percentage.
- Assignments Page: Table with Assignment Name, Subject, Due Date, and Status (Pending/Submitted).
- Notifications Panel: List of recent alerts with timestamps and category icons.
- Profile Page: Student photo, personal details, and editable fields for contact information.

Output in OpenProj:

- Added tasks: "Sketch Initial Designs" (Duration: 1 day) and "Create Digital Wireframes" (Duration: 3 days).
- Allocated time and resources; set dependency on Requirements Planning phase.

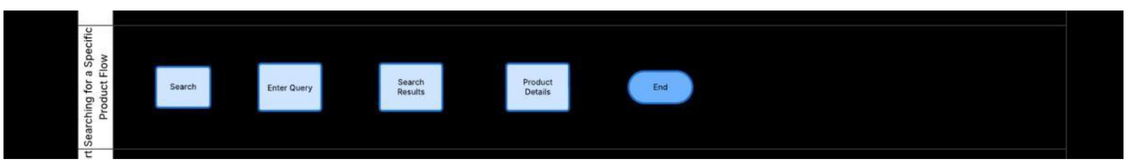

Fig 2: OpenProj – User Design Tasks

Step 3: Rapid Prototyping

The wireframes are converted into an interactive prototype using Axure RP. The prototype simulates the dashboard's navigation and interactive features, allowing stakeholders to experience the interface before development.

1. Develop Prototypes:

- Built a clickable prototype in Axure RP with navigation between Dashboard, Courses, Assignments, Notifications, and Profile pages.
- Added dynamic panels for notification badge updates and assignment status toggles.
- Simulated attendance progress bar animations and GPA calculation display.

2. Test Prototypes:

- Shared prototype via Axure Cloud link with faculty and student stakeholders.
- Feedback received: Users requested colour-coded assignment status (Green = Submitted, Red = Pending) and a quick-access floating button for adding reminders.
- Iterated on design: Applied colour coding, added reminder button, and improved sidebar icon labelling.

Output: Interactive prototype for the Student Dashboard with full navigation flow.

Output in OpenProj:

- Added tasks: "Develop Prototypes" (Duration: 3 days) and "Test Prototypes" (Duration: 2 days).
- Set milestones and dependencies on the User Design phase.

Fig 3: OpenProj – Rapid Prototyping Tasks

Step 4: User Acceptance / Testing

The completed prototype is reviewed by students and faculty. Usability testing is conducted to verify that the dashboard is intuitive, information is easy to locate, and all key actions can be completed efficiently.

1. Review Prototype:

- Conducted a walkthrough session with five students and two faculty members demonstrating the full dashboard flow.
- Positive feedback on layout clarity and navigation simplicity; minor issues noted with small text sizes on summary cards.

2. Conduct Usability Testing:

- Test tasks assigned: "Find today's schedule", "Mark an assignment as submitted", and "Check latest notification".
- All users completed tasks successfully within acceptable time limits.
- Documented feedback: Users requested larger card fonts, a dark mode option, and a search bar in the navigation.
- Updated prototype: Increased font sizes, added a dark/light mode toggle, and incorporated a global search bar.

Output: Documented usability test results with iterative design improvements applied.

Output in OpenProj:

- Added tasks: "Review Prototype" (Duration: 2 days) and "Usability Testing" (Duration: 3 days).
- Tracked progress and assigned review resources.

Fig 4: OpenProj – User Acceptance & Testing Tasks

Step 5: Implementation

The final step involves implementing the approved Student Dashboard design as a functional interface based on all collected feedback.

1. Develop Functional Interface:

- Implemented the dashboard using HTML, CSS (with responsive grid layout), and JavaScript.
- Built dynamic components: attendance progress bars, assignment status toggles, and notification badge counters.
- Applied the approved colour scheme (blue and white with accent colours), typography, and spacing from the final prototype.

2. Integrate Backend (if required):

- Connected the dashboard to a backend API to fetch real-time student data (attendance records, course list, assignment status).
- Integrated a notification service to push real-time alerts to the dashboard.
- Implemented secure session handling so only authenticated students can access their personalised dashboard.

Result

The experiment was successfully completed. The lifecycle stages of UI design using the RAD model were simulated for a Student Dashboard interface. Each stage — Requirements Planning, User Design, Rapid Prototyping, User Acceptance Testing, and Implementation — was documented and tracked using OpenProj. The Gantt charts confirm that all tasks were completed within the scheduled durations. The final prototype provides a fully interactive Student Dashboard that displays academic information clearly and enables key student actions efficiently.

Conclusion

The RAD model proved highly effective for developing the Student Dashboard UI. The iterative nature of the model allowed the design to be continuously refined through user feedback at every stage, resulting in an interface that closely matches real student needs. OpenProj enabled clear visualisation of the project timeline and helped in managing task dependencies and resource allocation. This experiment demonstrated how combining a structured development methodology with a dedicated project management tool leads to efficient and user-centred UI design outcomes.